Representing Outcomes of Experiments in Words, Symbols, Tables, and Graphs

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Outcome in Words

An **outcome in words** describes a result using ordinary language.

Example:
The die shows an even number.

Outcome in Symbols

An **outcome in symbols** uses mathematical symbols, numbers, or letters to describe a result.

Example:

$x = 4$ or $x \in \{2, 4, 6\}$

Words and Symbols Together

In Words	In Symbols
The die shows 4.	$x = 4$
The die shows an even number.	$x \in \{2, 4, 6\}$

Table

A **table** organizes outcomes and their corresponding information into rows and columns.

It helps us compare results and identify patterns.

Table Example

Outcome	Symbol
Heads	H
Tails	T

Each row shows one outcome.

Graph

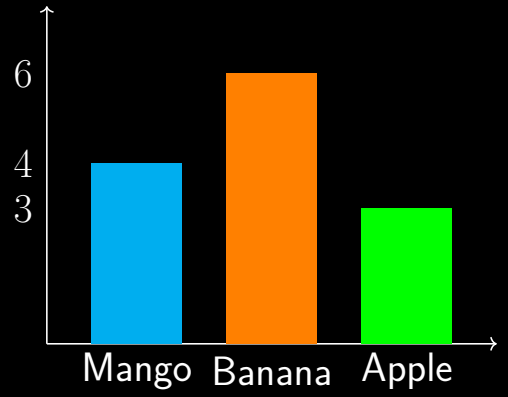
A **graph** presents information visually.

Common types include:

- Bar graph
- Line graph
- Pictograph

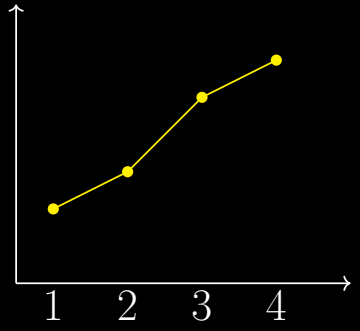
Bar Graph

A **bar graph** compares quantities in different categories.



Line Graph

A **line graph** shows changes in values over time or in an ordered sequence.



Pictograph

A **pictograph** uses pictures or symbols to represent quantities.

Pet	Number of Students
Dog	★★★
Cat	★★
Fish	★

Key: ★ $= 2$ students

Sample Space

A **sample space** is the set of all possible outcomes of an experiment.

Sample space when rolling a standard six-sided die:

$$S = \{1, 2, 3, 4, 5, 6\}$$

Frequency

The **frequency** of an outcome is the number of times it occurs.

Outcome	Frequency
Heads	6
Tails	4

Heads occurred 6 times.

Examples

Example 1

A coin is tossed once. Express the outcome of landing on heads.

In words: The coin shows heads.

In symbols: H

Example 2

A standard six-sided die is rolled once. Write the sample space. Then express the outcome of an odd number.

$$S = \{1, 2, 3, 4, 5, 6\}$$

In words: The die shows an odd number.

$$x \in \{1, 3, 5\}$$

Example 3

A coin is tossed 10 times. The coin lands on heads 6 times and on tails 4 times.

Outcome	Frequency
Heads	6
Tails	4

In words: Heads occurred more often than tails.

Practice Exercises

Practice 1

Express the outcomes as required.

A student tosses a coin once. List all possible outcomes in words and symbols, using H for heads and T for tails.

Practice 2

Express the outcomes as required.

A bag contains red, blue, and yellow balls. One ball is drawn at random. List the possible outcomes in words and represent them using set notation.

Practice 3

Express the outcomes as required.
The table shows the number of books read by a student over four weeks.

Week	Number of Books
1	2
2	4
3	3
4	5

Express the data in words and represent the number of books read using a line graph.

Practice 4

Express the outcomes as required.
A class records its students' preferred snacks. The results are shown below.

Snack	Number of Students
Bread	6
Biscuits	8
Fruit	5
Sandwich	7

Express the information in words and represent the data using a bar graph.

Practice 5

Express the outcomes as required.

The temperature recorded at noon for five days is shown below.

Day	Temperature (°C)
Monday	29
Tuesday	30
Wednesday	28
Thursday	31
Friday	32

Express the temperature data in words and represent the changes using a line graph.

Activities

Activity 1

Express the outcomes as required.

A standard six-sided die is rolled once. Write the sample space in symbols. Then express the outcomes that are less than 3 in words and symbols.

Activity 2

Express the outcomes as required.

A student records the number of tails obtained when tossing a coin 10 times. The results are shown below.

Number of Tails	Frequency
0	1
1	4
2	3
3	2

Express the outcomes in words and create a bar graph representing the frequency of each outcome.

Activity 3

Express the outcomes as required.

A spinner has four equal sections labeled P, Q, R, and S. Write the sample space in symbols. Then express the outcome of landing on a letter that comes after Q in alphabetical order, using words and symbols.

Activity 4

Express the outcomes as required.

A student rolls a die several times and records the following outcomes: $1, 3, 5, 1, 6, 3, 2, 1, 5, 3$.

Construct a frequency table showing each outcome and its frequency. Then express in words which outcome occurred most frequently.

Activity 5

Express the outcomes as required.

A school club surveys students about their favorite school subjects. The results are shown below.

Subject	Number of Students
Mathematics	11
Science	8
English	5
MAPEH	6

Write a statement describing the most and least popular subjects. Then represent the data using a suitable graph and label it properly.

Thank you!